

Introduction to C Programming

Programming Language

A programming language is a way to communicate instructions to a computer.

1. High-Level Language

- Easy for humans to read and write
- Uses simple English-like syntax
- Requires a compiler or interpreter to run

Examples:

- Python
- Java
- JavaScript

Example:

```
print("Hello")
```

2. Mid-Level Language

- Combines features of both high-level and low-level languages
- Provides better control over hardware
- Faster and more efficient

Examples:

- C
 - C++
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3. Low-Level Language

- Closer to machine hardware
- Difficult for humans to understand
- Very fast execution

Examples:

- Binary Language
- Assembly Language (ASM)

Example:

010111010

Program

A program is a set of instructions given to a computer to perform a specific task.

Example:

Input x

Input y

Add x and y

Print the result

Programming

Programming is the process of writing instructions (code) for a computer to solve problems or perform tasks.

Compiler

A compiler:

- Converts high-level language code into low-level machine code
- Translates the entire program at once
- Produces an output/executable file

Example Languages Using Compiler:

- C
 - C++
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Interpreter

An interpreter:

- Reads, translates, and executes code line by line
- Does not produce an output file
- Slower compared to a compiler

Example Languages Using Interpreter:

- Python
 - JavaScript
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IDE (Integrated Development Environment)

An IDE is software used for writing and developing programs.

Features of an IDE:

- Code Editor
- Terminal
- File Explorer
- Compiler
- Debugging Tools
- Auto Completion

Example:

- VS Code (Visual Studio Code)
- **Rules to name a C FILE:**
 - **File name should not contain any white space:**
My file.c first code.c
 - **File name should not start with number:**
5code.c 2mycode.c
 - **File name must end with .c extension**
Code.c first.c